F78AB - Actuarial and Financial Mathematics B

Tutorial 3 – Real interest rates and index-linked securities.

Question 1 A bond with the following characteristics was issued in November 2008.

Term	5 years
Coupon	4.5 % p.a.
Coupon payment frequency	Annually
Issue price	100 %
Redemption rate	105 %

Calculate the real yield-to-redemption, with reference to the RPI, to an investor not subject to tax.

Question 2 [Relationship between real and money yields (I).]

1. An investor pays £100 for an asset. Exactly one year later, the investor sells the asset for £110. Over the same year, the RPI has increased by the rate 3%.

   (a) Calculate the investor's money yield.

   (b) Calculate the investor's real yield.

2. An investor pays £1 for an asset. Exactly one year later, the investor sells the asset for $£(1+i)$, with $i > -1$. Over the same year, the RPI has increased at the rate q.

   Show that the real yield i' on the transaction satisfies

$$\boxed{1+i' = \frac{1+i}{1+q}}$$

Question 3 [Relationship between real and money yields (II).] Assume that time is measured in years. At time 0, an investor pays £100 for an investment which produces a cashflow of £50 at time 1 and a cashflow of £60 at time 2. There are no further cashflows or value from the investment.

1. Calculate the investor's annualised money yield i.

2. Calculate the investor's annualised real yield i', based on the Inflation Index below.

Time	0	1	2
Inflation Index at time	100	104	112

3. Calculate the average annualised rate of change $\bar{q}$ in the Inflation Index.

4. Explain why, in general,

$$1+i' \neq \frac{1+i}{1+\bar{q}}\,.$$

5. How must the inflation rate change as a function of time to have the equality $1+i' = \frac{1+i}{1+\bar{q}}$?

Question 4 A particular transaction will provide an investor with a yield of 6.400% per annum.

1. Assuming that over the period of the transaction the annual rate of inflation will be constant and equal to 2.800%, calculate the investor's real yield on the transaction.

2. Calculate the possible values of the constant annual rate of inflation q such that the investor obtains a real yield on the transaction of at least 3.750% per annum.

Question 5 On 14 July 2006 a government stock was issued with a term of three years. The issue-price of the stock was 95% and redemption was at par. Dividends were payable annually in arrears with a coupon rate of 7.6% per annum. An investor who is not liable to tax purchased some of the stock on the issue date and held it until redemption.

1. Find the yield obtained by the investor. [NOTE: You should generally assume that "yield" means "money yield".]

2. Using the inflation index below, find the real yield on the investment.

Year	2006	2007	2008	2009
Inflation index on 14 July of Year	123.6	127.9	135.2	139.3

Question 6 A loan of £25 000 was issued in February 2010 and was repaid at par after three years. Interest accrues on the loan at the rate of 8% per annum and is payable annually in arrears.

The value of an Inflation Index at various times was as follows.

Date	Feb 2010	Feb 2011	Feb 2012	Feb 2013
Inflation index at date	205.0	215.6	223.5	231.5

Calculate the real rate of return on the loan by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Question 7 On 1 December 2009, Highland Life, an insurance company, issued a bond with a term to redemption of 2 years and a coupon rate of 6% p.a. Coupons were paid annually in arrears, and the bond was redeemed at par. An investor purchased £1 000 nominal of this bond and held it until redemption.

The Scottish Consumer Price Index (SCPI) had the following values:

Date	Dec 2009	Dec 2010	Dec 2011
SCPI(Date)	150	155	160

The investor obtained a real yield of 4% p.a. effective, based on the SCPI. Calculate the price that she paid on 1 December 2009.

Question 8 An ordinary share pays dividends on each 31 December. A dividend of 35p per share was paid on 31 December 2011. The dividend growth is expected to be 3% in 2012, and a further 5% in 2013. Thereafter, dividends are expected to grow at 6% per annum compound in perpetuity.

1. Calculate the present value of the dividend stream described above at a rate of interest of 8% per annum effective for an investor holding 100 shares on 1 January 2012.

An investor buys 100 shares for £17.20 each on 1 January 2012. He expects to sell the shares for £18 on 1 January 2015.

Assume that dividends grow as expected and use the following values of the inflation index:

Year	2012	2013	2014	2015
Inflation index at start of year:	110.0	112.3	113.2	113.8

2. Calculate the investor's expected real rate of return.

Question 9 An individual has invested a sum of £10m. Exactly one year later, the investment is worth £11.1m. An index of prices has a value of 112 at the beginning of the investment and 120 at the end of the investment. The investor pays tax at 40% on all money returns from investment. Calculate:

1. The money rate of return per annum before tax.

2. The rate of inflation.

3. The real rate of return per annum after tax.

Question 10 A tax advisor is assisting a client in choosing between three types of investment. The client pays tax at 40% on income and 40% on capital gains.

Investment A requires the investment of £1m and provides an income of £0.1m per year in arrears for ten years. Income tax is deducted at source. At the end of the ten years, the investment of £1m is returned.

In Investment B, the initial sum of £1m accumulates at the rate of 10% per annum compound for ten years. At the end of the ten years, the accumulated value of the investment is returned to the investor after deduction of capital gains tax.

Investment C is identical to Investment B except that the initial sum is deemed, for tax purposes, to have increased in line with the index of consumer prices between the date of the investment and the end of the ten-year period. The index of consumer prices is expected to increase by 4% per annum compound over the period.

1. Calculate the net rate of return expected from each of the investments.

2. Explain why the expected rate of return is higher for Investment C than for Investment B and is higher for Investment B than for Investment A.

Question 11 State the characteristics of index-linked government bonds.

Question 12 On 9 October 2007 an investor, who is not liable to tax, had the choice of purchasing either:

(A) $7\frac{1}{2}\%$ Treasury Stock at price £107 per £100 nominal repayable at par.

(B) 2% Index-Linked Treasury Stock 2016 at a price of £207 per £100 nominal.

The RPI base figure for indexing is 69.5 and the next coupon is to be paid on 9 April 2008. The time lag for inflation indexing is 8 months.

Both stocks pay interest half-yearly in arrears and are redeemed on 9 October 2016. The latest published value of the RPI is 207.3, which is the RPI for August 2007.

Assuming that today is 9 October 2007 and the RPI will grow continuously at a rate of 2.5% per annum from its latest published value,

1. Verify that the real yield from stock (A) as at 9 October 2007 is 4% per annum effective.

2. By considering the series of future receipts from stock (B), derive a formula for the present real value of £100 nominal of that stock.

3. Calculate the real yield as at 9 October 2007 from stock (B) by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Question 13 On 15 March 2006 the government of a country issued an index-linked bond of term 6 years. Coupons are payable half-yearly in arrears, and the annual nominal coupon rate is 3%.

Interest and capital payments are indexed by reference to the value of an inflation index with a time lag of 8 months.

A tax-exempt investor purchased the stock at £111 per £100 nominal on 16 September 2009, just after the coupon payment had been made.

You are given the following values of the inflation index:

Date	Jul 2005	Mar 2006	Jul 2009	Sep 2009
Inflation Index (Date)	110.5	112.1	126.7	127.4

1. Calculate the amount of the coupon payment per £100 nominal stock on 15 March 2010.

2. Calculate the effective real annual yield to the investor on 16 September 2009. You should assume that the inflation index will increase at a constant rate of 4% per annum effective from its value in September 2009.

3. Without doing any further calculations, explain how the effective real annual yield would alter, if at all, if the inflation index for July 2005 had been more than 110.5 and all the other index values were unchanged.

Solutions: Q1. 2.22% p.a.; Q2.1. 10%; Q2.1. 6.796%; Q3.1. 6.394%; Q3.2. 1.077%; Q3.3. 5.830%; Q4.1. 3.502%; Q4.2. $q \leq 2.554\%$; Q5.1 9.596%; Q5.2 5.304%; Q6. 3.670%; Q7. £974.61; Q8.1. £1785.81; Q8.2. 2.523%; Q9.1. 11%; Q9.2. 7.143%; Q9.3. -0.507%; Q10.1. A: 6% p.a, B: 6.941% p.a, C: 7.947% p.a; Q9.2. 7.143%; Q12.3. 6.481%; Q13.1. £1.71991; Q13.2. 3.52% p.a.

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Solutions to Tutorial 3 – Real interest rates and index-linked securities.

Solution 1

Real cashflows are calculated as below:

Time t (years)	0	1	2	3	4	5
Money cashflow	-100	4.5	4.5	4.5	4.5	109.5
Payment date	Nov 2008	Nov 2009	Nov 2010	Nov 2011	Nov 2012	Nov 2013
RPI(Payment date)	216.0	216.6	226.8	238.5	245.6	252.1
Real cashflow at Nov 2008	$-100 \times \frac{216.0}{216.0}$ $= -100$	$4.5 \times \frac{216.0}{216.6}$ $= 4.488$	$4.5 \times \frac{216.0}{226.8}$ $= 4.286$	$4.5 \times \frac{216.0}{238.5}$ $= 4.075$	$4.5 \times \frac{216.0}{245.6}$ $= 3.958$	$109.5 \times \frac{216.0}{252.1}$ $= 93.820$

Denote the annual real yield by i'. Defining

$$f(j) := 100(1+j)^5 - 4.488(1+j)^4 - 4.286(1+j)^3 - 4.075(1+j)^2 - 3.958(1+j) - 93.820,$$

we seek to find i' that satisfies $f(i') = 0$.

By trial and error, we find $f(0.020) = 1.095$ and $f(0.025) = -1.413$. Thus i' lies between 0.020 and 0.025. Then linear interpolation gives $i' \approx 0.022$.

As a check, $f(0.022) = 0.098$; not great but acceptable. If we wished to get a more precise answer, we could do a second round of linear interpolation, e.g. using $f(0.022) = 0.098$ and $f(0.024) = -0.907$. This results in a more precise value of $i' = 2.22\%$. As a check, $f(0.0222) = -0.002$; better!

Solution 2

1. (a) The investor's money yield is $\frac{110}{100} - 1 = 10\%$.

   (b) Expressing the cashflow of £110 in terms of purchasing power at the start of the transaction year, its real value is $\frac{110}{1.03} = 106.796$. Using i' to denote the real yield, the equation of value is

   $$100(1+i') - 106.796 = 0 \quad \Rightarrow \quad i' = \frac{106.796}{100} - 1 = 0.06796,$$

   i.e. the real yield is 6.796%.

2. Expressing the cashflow of £$(1+i)$ in terms of purchasing power at the start of the transaction year, its real value is $\frac{1+i}{1+q}$. Using i' to denote the real yield, the equation of value is

   $$1 \times (1+i') - \frac{1+i}{1+q} = 0 \quad \Rightarrow \quad i' = \frac{1+i}{1+q} - 1.$$

Solution 3

1. The investor's money yield i is the solution to
$$100(1+i)^2 - 50(1+i) - 60 = 0. \tag{1}$$
Solving as a quadratic equation, we find $i = 6.394\%$.

2. Expressing the cashflow of £50 at time 1 in terms of purchasing power at time 0, its real value is $\frac{50}{104} \times 100 = 48.077$. Expressing the cashflow of £60 at time 2 in terms of purchasing power at time 0, its real value is $\frac{60}{112} \times 100 = 53.571$.
Using i' to denote the real yield, the equation of value is
$$100(1+i')^2 - 48.077(1+i') - 53.571 = 0. \tag{2}$$
Solving as a quadratic equation, we find $i' = 1.077\%$.

3. The average rate of change $\bar{q}$ of the Inflation Index satisfies
$$1 + \bar{q} = \left(\frac{112}{100}\right)^{1/2} \qquad \Rightarrow \qquad \bar{q} = 0.05830.$$

4. For each money cashflow, we use the actual inflation rate at the time of the cashflow to convert it to a real value. Using the average inflation rate would result, in general, in a different value. Thus we cannot, in general, calculate the real yield directly from the money yield and the average inflation rate.

5. We have the equality $(1+i') = (1+i)/(1+\bar{q})$ when the inflation rate is constant across time, and/or in the simple case where we have only two time points at which cashflows occur (as in Question 2).

Solution 4 When the annual rate of inflation is constant, then the real yield i' and the money yield i are linked by the equation
$$1 + i' = \frac{1+i}{1+q}, \tag{3}$$
in which q is the constant annual rate of inflation.

1. $i' = 1.0640/1.0280 - 1 = 0.03502$, i.e. the real yield is 3.502% per annum.

2. We are asked to have $1 + i' \geq 1.03750$. Using (3) and with $i > -1$ and $q > -1$, we find
$$\frac{1.0640}{1+q} \geq 1.03750 \quad \Rightarrow \quad q \leq 0.02554$$
i.e. as long as the constant annual inflation rate is less than 2.554% per annum, the investor's real yield will exceed 3.75% per annum.

Solution 5

1. Denote the annual money yield by i. The equation of value for the money cashflows at time $t = 3$ is
$$95(1+i)^3 - 7.6(1+i)^2 - 7.6(1+i) - 107.6 = 0.$$
By trial and error, we find that the solution i lies between 9.5% and 10%. Then linear interpolation gives $i \approx 9.596\%$.
As a check: $95(1.09596)^3 - 7.6(1.09596)^2 - 7.6(1.09596) - 107.6 = -0.001$. Acceptable.

2.

Time t	0	1	2	3
Cashflow	-95	7.6	7.6	107.6
Payment date	14 Jul 2006	14 Jul 2007	14 Jul 2008	14 Jul 2009
Index(Payment date)	123.6	127.9	135.2	139.3
Real cash flows	-95	$\frac{123.6}{127.9} \times 7.6$	$\frac{123.6}{135.2} \times 7.6$	$\frac{123.6}{139.3} \times 107.6$
	$= -95$	$= 7.344$	$= 6.948$	$= 95.473$

The equation of value for the real cashflows at $t = 3$ is

$$95(1+i')^3 - 7.344(1+i')^2 - 6.948(1+i') - 95.473 = 0.$$

By trial and error, we find that the solution i' lies between 5% and 5.5%. Then linear interpolation gives $i' \approx 5.304\%$.

As a check: $95(1.05304)^3 - 7.344(1.05304)^2 - 6.948(1.05304) - 95.473 = -0.001$. Acceptable.

Solution 6

The amount of interest paid at end of each year is $0.08 \times 25\,000 = 2\,000$. The real rate of return i' solves the equation of value

$$\begin{aligned} f(j) =& 25\,000(1+j)^3 - 2\,000\frac{205.0}{215.6}(1+j)^2 - 2\,000\frac{205.0}{223.5}(1+j) - 27\,000\frac{205.0}{231.5} \\ =& 25\,000(1+j)^3 - 1\,901.67(1+j)^2 - 1\,834.45(1+j) - 23\,909.29. \end{aligned}$$

at $f(i') = 0$.

First estimate of i': the inflation index has increased over the three years by the factor $231.5/205 = 1.129$, which gives an average inflation rate of 4.1% p.a. As the money yield is 8% p.a. and the annual increases in the inflation index haven't been too volatile, then we would expect the real yield to be not too far from $\frac{1.08}{1.041} - 1 = 3.75\%$ p.a.

By trial and error, we find that the solution i' lies between 3.5% and 4%. Then linear interpolation gives $i' \approx 0.03670$.

As a check, $f(0.03670) = -0.126$: acceptable given that we are working in £000s and not £00s.

Solution 7 The price P per £1 000 nominal is

$$P = 60 \times \frac{150}{155} \times (1.04)^{-1} + 1060 \times \frac{150}{160} \times (1.04)^{-2} = 974.61.$$

Solution 8

1. The present value at 1 January 2012 of the future stream of dividends equals

$$\begin{aligned}\text{PV} =& 100 \times 0.35 \times \left(\frac{1.03}{1.08} + \frac{1.03 \times 1.05}{1.08^2} + \frac{1.03 \times 1.05 \times 1.06}{1.08^3} + \frac{1.03 \times 1.05 \times 1.06^2}{1.08^4} + \cdots \right) \\ =& 35 \times \left(\frac{1.03}{1.08} + \frac{1.03 \times 1.05}{1.08^2} + \frac{1.03 \times 1.05 \times 1.06}{1.08^3} \sum_{n=0}^{\infty} \left(\frac{1.06}{1.08} \right)^n \right).\end{aligned}$$

   As

$$\sum_{n=0}^{\infty} \left(\frac{1.06}{1.08} \right)^n = \frac{1}{1 - \frac{1.06}{1.08}} = 54,$$

   we find

$$\begin{aligned}\text{PV} =& 35 \times \left(\frac{1.03}{1.08} + \frac{1.03 \times 1.05}{1.08^2} + \frac{1.03 \times 1.05 \times 1.06}{1.08^3} \times 54 \right) \\ =& 35 \times (0.95370 + 0.92721 + 49.14223) \\ =& 1\,785.81.\end{aligned}$$

2. Denote by i' the annual real rate of return. The equation of value is

$$\begin{aligned}f(j) :=& \, 1\,720 - \frac{35 \times 1.03}{112.3} \times 110v - \frac{35 \times 1.03 \times 1.05}{113.2} \times 110v^2 \\ & - \frac{35 \times 1.03 \times 1.05 \times 1.06}{113.8} \times 110v^3 - \frac{1\,800}{113.8} 110v^3 \\ =& \, 1\,720 - 35.312v - 36.782v^2 - 1778.678v^3.\end{aligned}$$

   For an initial estimate of i', assume that all income received at end of 3 years, i.e.

$$1720 \approx 1850.772v^3.$$

   This gives $v^3 \approx 0.97587 \Rightarrow i' \approx 2.5\%$.

   As $f(0.025) = -1.140$ and $f(0.03) = 23.304$, linear interpolation gives $i' = 2.523\%$.

   As a check, $f(0.02523) = -0.0049$: acceptable.

Solution 9

1. The money rate of return is i where $(1 + i) = 11.1/10$, i.e. $i = 11\%$.

2. The rate of inflation is q where $(1 + q) = 120/112$, i.e. $q = 7.143\%$.

3. The investor keeps 60% of the monetary investment return £1.1 million after paying tax, i.e. the after-tax monetary investment return is £0.66 million. Thus the value of the investment at the end of the year is £$(10 + 0.66)$ = £10.66 million.

   The net real rate of return per annum is i' satisfies the equation of value

$$10(1 + i') - \frac{10.66}{1 + q} = 0 \quad \Rightarrow \quad i' = -0.005068.$$

Solution 10

1. Investment A: the net return is 6% per annum effective.

   Investment B: the investment will accumulate to £1m $\times 1.1^{10} = £2.59374$m at the end of the ten years. The taxable capital gain is $2.59374 - 1 = £1.59374$m. The capital gains tax paid is $1.59374 \times 40\% = 0.63750$ to leave a net accumulated investment value of $£(2.59374 - 0.63750) = 1.95624$m. Thus the equation of value is $(1 + i_B)^{10} = 1.95624$ $\Rightarrow i_B = 6.941\%$.

   Investment C: again the investment will accumulate to £2.59374m at the end of ten years. However, the indexed purchase price is subtracted from the accumulated value of the investment in order to calculate the taxable capital gain, i.e.

$$\text{Taxable capital gain is } 2.59374 - 1 \times 1.04^{10} = 1.11350$$
$$\Rightarrow \text{Capital gains tax paid } = 1.11350 \times 40\% = 0.44540\text{m}.$$

   Thus the net accumulated investment value is $£(2.59374 - 0.44540) = 2.14834$m. Thus the equation of value is $(1 + i_C)^{10} = 2.14834 \Rightarrow i_C = 7.947\%$.

2. All investments give a gross return of 10% per annum effective.

   Investment B gives a higher return than A because the tax is deferred until the end of the investment period, since capital gains tax is paid and not income tax.

   Investment C gives a higher return than investment B because the tax is only paid on the real return over the ten year period which is lower than the money return.

Solution 11

1. Payments guaranteed by government.
2. Low default risk.
3. Can be various different indexation provisions but, in general, protection is given against a fall in the purchasing power of money.
4. Fairly liquid (i.e. large issue size and ability to deal in large quantities) compared with corporate issues, but not compared with conventional issues.
5. Normally coupon and capital payments both indexed to increases in a given price index with a lag.
6. Low volatility of return and low expected real return.
7. More or less guaranteed real return if held to maturity (can vary due to indexation lag).
8. Money return is not guaranteed.

Solution 12

1. As the inflation index is assumed to grow at a constant rate of 2.5% p.a. we can use the fact that the real yield i' and the money yield i are related by

$$1 + i' = \frac{1 + i}{1.025}.$$

   to do the verification. If the real yield i'_A for stock A is 4% p.a. then the money yield of A must be $i_A = (1 + i'_A)(1.025) - 1 = 0.066$ p.a. As

$$107 - 7.5a^{(2)}_{\overline{9}|} - 100v^9 = -0.22814 \quad \text{at } i_A = 0.066,$$

   this means that the money yield is 6.6% p.a, which verifies that the real yield is 4% p.a.

2. As the RPI at the base date is 69.5, we estimate the money payments per £100 nominal from the index-linked stock as follows.

Time t	0.5	1	1.5	$\cdots$	9
Date at time t	9 Apr 08	9 Oct 08	9 Apr 09	$\cdots$	9 Oct 16
Lagged RPI date	Aug 07	Feb 08	Aug 08	$\cdots$	Feb 16
RPI(Lagged date)	207.3	$207.3 \times 1.025^{\frac{1}{2}}$	207.3×1.025^{1}	$\cdots$	$207.3 \times 1.025^{8\frac{1}{2}}$
Money payment	$\frac{207.3}{69.5}$	$\frac{207.3\times 1.025^{\frac{1}{2}}}{69.5}$	$\frac{207.3\times 1.025^{1}}{69.5}$	$\cdots$	$101 \times \frac{207.3\times 1.025^{8\frac{1}{2}}}{69.5}$

The real values calculated at 9 October 2007 can be found as follows. As the projected RPI at October 2007 is $207.3 \times 1.025^{\frac{2}{12}}$,

Time t	0.5	1	1.5	$\cdots$
Money payment	$\frac{207.3}{69.5}$	$\frac{207.3\times 1.025^{\frac{1}{2}}}{69.5}$	$\frac{207.3\times 1.025^{1}}{69.5}$	$\cdots$
Payment date	9 Apr 08	9 Oct 08	9 Apr 09	$\cdots$
RPI(Payment date)	$207.3 \times 1.025^{\frac{8}{12}}$	$207.3 \times 1.025^{1\frac{2}{12}}$	$207.3 \times 1.025^{1\frac{8}{12}}$	$\cdots$
Real payment	$\frac{207.3}{69.5} \times \frac{207.3\times 1.025^{\frac{2}{12}}}{207.3\times 1.025^{\frac{8}{12}}}$	$\frac{207.3\times 1.025^{\frac{1}{2}}}{69.5} \times \frac{207.3\times 1.025^{\frac{2}{12}}}{207.3\times 1.025^{1\frac{2}{12}}}$	$\frac{207.3\times 1.025^{1}}{69.5} \times \frac{207.3\times 1.025^{\frac{2}{12}}}{207.3\times 1.025^{1\frac{8}{12}}}$	$\cdots$
at Oct 2007	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$\cdots$

with the total real value of the redemption payment and last coupon payment being $101 \times \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$.

Hence if i'_B is the real rate of return, the formula for the present real value of £100 nominal of stock (B) is:

$$\frac{207.3}{69.5(1.025)^{1/2}} \left(2a_{\overline{9}|}^{(2)i'_B} + 100(1+i'_B)^{-9}\right). \tag{4}$$

3. Having found the expression (4), the real yield is most easily found by solving the equation of value formulated at $t = 0$, that is, on 9 October 2007. We express it as follows:

$$207 = \frac{207.3}{69.5(1.025)^{1/2}} \left(2a_{\overline{9}|}^{(2)i'_B} + 100(1+i'_B)^{-9}\right).$$

By trial and error, we find the solution lies between 0.06 and 0.065. Then linear interpolation gives $i'_B \approx 0.06481$.

Solution 13

1. As the base date for indexing is July 2005, the coupon will be:

$$\frac{0.03 \times 100}{2} \times \frac{\text{II(Jul 2009)}}{\text{II(Jul 2005)}} = 1.5 \times \frac{126.7}{110.5} = \text{£}1.71991 \text{ per £100 nominal.}$$

2. As the RPI at the base date is 110.5, we estimate the money payments per £100 nominal from the index-linked stock as follows.

Time t	0.5	1	1.5	2	2.5
Date at time t	Mar 10	Sep 10	Mar 11	Sep 11	Mar 12
Lagged RPI date	Jul 09	Jan 10	Jul 10	Jan 11	Jul 11
RPI(Lagged date)	126.7	$127.4 \times 1.04^{\frac{4}{12}}$	$127.4 \times 1.04^{\frac{10}{12}}$	$127.4 \times 1.04^{1\frac{4}{12}}$	$127.4 \times 1.04^{1\frac{10}{12}}$
Money payment	$1.5\frac{126.7}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{4}{12}}}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{10}{12}}}{110.5}$	$1.5\frac{127.4\times1.04^{1\frac{4}{12}}}{110.5}$	$101.5\frac{127.4\times1.04^{1\frac{10}{12}}}{110.5}$

The real values are calculated at September 2009, at which date the RPI is 127.4.

Time t	0.5	1	1.5	$\cdots$
Money payment	$1.5\frac{126.7}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{4}{12}}}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{10}{12}}}{110.5}$	$\cdots$
Payment date	Mar 10	Sep 10	Mar 11	$\cdots$
RPI(Payment date)	$127.4(1.04)^{0.5}$	$127.4(1.04)^{1}$	$127.4(1.04)^{1.5}$	$\cdots$
Real payment	$1.5\frac{126.7}{110.5}\frac{127.4}{127.4(1.04)^{0.5}}$	$1.5\frac{127.4\times1.04^{\frac{4}{12}}}{110.5}\frac{127.4}{127.4(1.04)^{1}}$	$1.5\frac{127.4\times1.04^{\frac{10}{12}}}{110.5}\frac{127.4}{127.4(1.04)^{1.5}}$	$\cdots$
at Sep 2009	$= 1.5\frac{126.7}{110.5}\frac{1}{(1.04)^{0.5}}$	$= 1.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}}$	$= 1.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}}$	$\cdots$
	$= 1.68651$	$= 1.68478$	$= 1.68478$	$\cdots$

with the total real value of the redemption payment and last coupon payment being

$$101.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}} = 114.00336.$$

Using time units equal to 6 months, the equation of real value is

$$f(j) := 111(1+j)^5 - 1.68651(1+j)^4 - 1.68478(1+j)^3 - 1.68478(1+j)^2 - 1.68478(1+j)^1 - 114.00336.$$

As a first estimate of the solution to $f(j^\star) = 0$, assume that the all the income is received at the end of the investment. Then $111(1+j)^5 \approx 120.74421$, resulting in a first estimate of $j^\star \approx 1.7\%$.

As $f(0.015) = -1.422$ and $f(0.02) = 1.465$, linear interpolation gives $j^\star \approx 1.746\%$. Hence the annual real yield is 3.52%.

3. The real yield decreases. If the inflation index at July 2005 had been greater than 110.5, then the money payments would decrease. Consequently, so would the real payments. Thus the investor would be paying the same amount £111 for less and hence the investor's yield would be less than 3.52% p.a.